

The Art & Science of Commercial Due Diligence: A Data Driven Approach

Workshop

12.03.2025

KEARNEY



Case Study: Forecasting newspaper sales

You were staffed on a project for a client who commissioned **due diligence** on a newspaper publisher - **PaperRush**



Market situation

- For several years, the market has been experiencing a declining **trend of -10% year over year**.
- Retailers' margins are about 15% of the newspaper's selling price.
- There is a significant **difference between production costs and revenue from sales**, which currently amounts to approximately five times the production cost.
- It is common practice to supply stores with significantly more copies of newspapers than actual demand requires **to avoid stockouts**.



Task and deliverables

- Your task is to verify whether it is possible to develop a model that can accurately predict the sales of each title in every store and whether implementing this model **will improve the company's profitability**.
- You need to prepare a **report outlining the current state of the company**, including newspaper sales and deliveries in previous years, the current financial situation, the sales distribution of individual titles, and the performance of different retail chains and store types. Additionally, the report should include an analysis of **how implementing your model will impact the company's outlook**.



Action

- To accurately model sales, you need to understand the client's product portfolio and identify which product and store characteristics influence sales levels.
- You will need to integrate data from various sources, conduct a feature importance analysis, apply appropriate models for different product groups, evaluate prediction accuracy, and estimate the financial impact.

There are two types of products that require different approaches to modeling

1 Weeklies

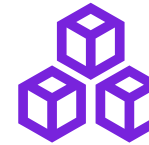
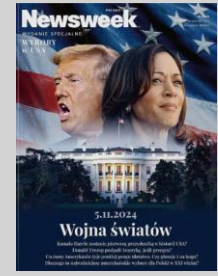
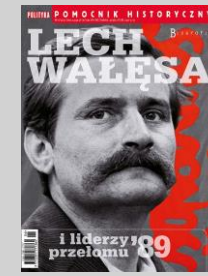
Opportunities

- Regular sales over times
- Low seasonality
- Issued week after week

Challenges:

- Content plays a secondary role
- One title published every two weeks

Examples:



Opportunities

- Content and segment play a key roles
- Individual segments may have a stable sales history in the store

Challenges:

- Irregular sales over times
- Each issue is basically a new product

Examples:



SIPs 2

To build the model
you received
outputs from the
PaperRush
database

FACT_SALES

Product ID
Store ID
SoldQty
DrawQty



DIM_PRODUCT

Product ID
Barcode
Title
Type (Weeklies / SIP)
On Sale Date
Off Sale Date
Price
Segment
Sub-Segment
Frequency



DIM_STORE

Store ID
Postal Code
Chain
City
Region
Class of trade
Merchandised
Facings
Pockets



DEMOGRAPHICS

Postal Code
Age groups
Race
Education
Household Income
Household Type

Share of population



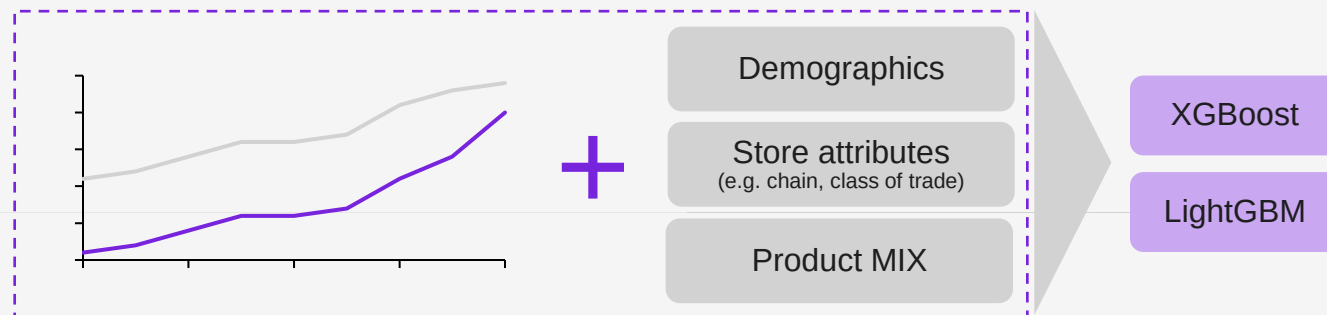
CONTENTS_EMB

Product ID
Content Embedding

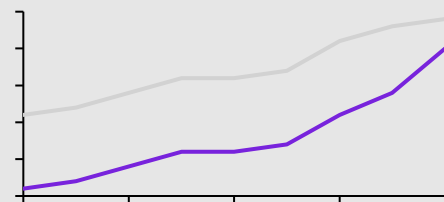


This problem
requires a holistic
approach

Store Performance



Weeklies Performance



ARIMA

LSTM

Prophet



ABC

Special Interest Publications Performance



SIP Similarity

Historical Sales

XGBoost

LightGBM

When solving a business case, you will use a framework from real due diligence projects

Conduct an exploratory analysis and build a forecasting model



1. Understand the data: calculate the correlation between the variables, perform feature importance, build a simple model and evaluate its accuracy
2. Use the ensemble approach to build a demand forecasting model



Carry out a pilot and predict future sales based on printing schedule



3. Calculate sales predictions for the provided printing schedule for each store and product combinations - this will be one month's issues for a subset of all stores
4. In addition to the sale itself, suggest the number of copies delivered. **Remember that your goal is to maximize the profit, and any stockout generates losses**
5. Upload your predictions and recommended draw **at least 2 weeks before final presentations¹**



Outline the financial benefits of model implementation



6. 2 days after sending the recommendation we will provide you with real sales values
7. Calculate the financial benefits generated by the one-month pilot and extrapolate them to annual values
8. **Present the current financial situation of the company and showcase your approach along with the benefits of implementing your model**



Thank you

Grzegorz Krochmal
Senior Analytics Consultant

Barbara Blicharz
Data Engineer

Jakub Filipczyk
Data Engineer

Business Case point of contact
jakub.filipczyk@kearney.com

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